

0	7	.	1
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What is a tumour?

[2 marks]

0	7	.	2
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Describe how you would determine a **reliable** mitotic index (MI) from tissue observed with an optical microscope.Do **not** include details of how you would prepare the tissue observed with an optical microscope.**[3 marks]**

Question 7 continues on the next page**Turn over ►**

Tumours detected under the skin can be a symptom of cancer.
Scientists investigated the link between the MI of tumours and skin cancer in dogs.

They found the MI of tumours in many dogs and recorded:

1. the tumour grade
2. the median survival time after the tumour is detected.

Tumour grade can be measured using this scale:

- grade 1 – low level cancer
- grade 2 – medium level cancer
- grade 3 – high level cancer.

The scientists used a statistical test to calculate the probability (P) of the difference between median survival time in dogs with MI < 5 and dogs with MI > 5 being caused by chance.

Figure 8 and **Table 4** show the scientists' results.

Figure 8

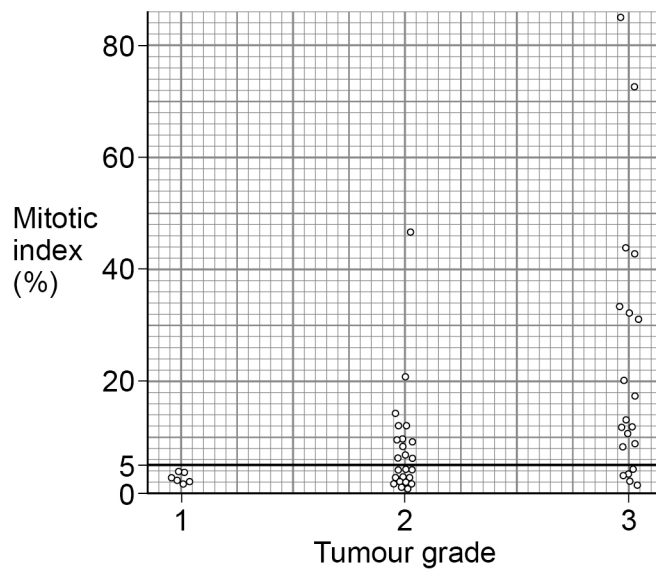


Table 4

Mitotic index (%)	Median survival time / months	Probability (P value)
< 5	70	< 0.001
> 5	2	

Key

◦ represents one dog

[4 marks]

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